

Reference: 大黄书, Chapter 4 & 2208.05180 & Ginsparg

Conformal Transformation:

an invertible mapping satisfying: $g_{\mu\nu}(x) \rightarrow g'_{\mu\nu}(x) = \Lambda(x) g_{\mu\nu}(x)$
 preserve angle, locally "rotation" + "dilation"

If the transformation is infinitely small: $x'^{\mu} = x^{\mu} + \epsilon^{\mu}$

$$g'_{\mu\nu} = g_{\mu\nu} - (\partial_{\mu} \epsilon_{\nu} + \partial_{\nu} \epsilon_{\mu}) = f(x) g_{\mu\nu}$$

Then we have the properties:

1) [Trace] $f(x) = \frac{2}{d} \partial_{\rho} \epsilon^{\rho}$

Take $g_{\mu\nu} = \eta_{\mu\nu}$,

2) [Partial + Permute]: $2 \partial_{\mu} \partial_{\nu} \epsilon_{\rho} = \eta_{\mu\rho} \partial_{\nu} f + \eta_{\nu\rho} \partial_{\mu} f - \eta_{\mu\nu} \partial_{\rho} f$

3) [contract $\eta^{\mu\nu}$]: $2 \partial^2 \epsilon_{\mu} = (2-d) \partial_{\mu} f$

4) [$\partial^2(\text{original}) = \partial^{\nu}(z)$]: $(2-d) \partial_{\mu} \partial_{\nu} f = 2 \partial_{\rho} \partial^{\rho} \partial_{\nu} \epsilon_{\mu} = \eta_{\mu\nu} \partial^2 f$

5) [contract $\eta^{\mu\nu}$]: $(d-1) \partial^2 f = 0$

if $d=1$: (5) means no constrain on f

if $d \geq 3$:

(4)+(5) imply $\partial_{\mu} \partial_{\nu} f = 0 \rightarrow f = A + B_{\mu} x^{\mu}$

(2) imply $\epsilon_{\mu} = a_{\mu} + b_{\mu\nu} x^{\nu} + c_{\mu\nu\rho} x^{\nu} x^{\rho}$

original + (1) gives $b_{\mu\nu} + b_{\nu\mu} = \frac{2}{d} b^2 \lambda \eta_{\mu\nu} \rightarrow b_{\mu\nu} = \underbrace{\text{antisymmetric}}_{\text{rotation}} + \underbrace{\text{trace}}_{\text{global dialation}}$

(2) imply $c_{\mu\nu\rho} = \eta_{\mu\rho} b_{\nu} + \eta_{\mu\nu} b_{\rho} - \eta_{\nu\rho} b_{\mu}$ ($b_{\mu} \equiv \frac{1}{d} c^{\sigma}{}_{\sigma\mu}$)

$\therefore \epsilon'_{\mu} = c_{\mu\nu\rho} x^{\nu} x^{\rho} = 2(\vec{b} \cdot \vec{x}) x_{\mu} - x^2 b_{\mu} \rightarrow \text{special conformal transformation}$

For finite transformation:

translation $x'^{\mu} = x^{\mu} + a^{\mu}$

dilation $x'^{\mu} = \alpha x^{\mu}$

rigid rotation $x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$

SCT $x'^{\mu} = \frac{x^{\mu} - b^{\mu} x^2}{1 - 2\vec{b} \cdot \vec{x} + b^2 x^2}$ or $\frac{x'^{\mu}}{x'^2} = \frac{x^{\mu}}{x^2} - b^{\mu}$

easy to verify, see stackexchange "SCT"!

for SCT: $x'^2 = \frac{x^2}{1 - 2\vec{b} \cdot \vec{x} + b^2 x^2} = \frac{x^2}{\Lambda^2}$

Generators: $iG_a \Phi = \frac{\delta x^{\mu}}{\delta \omega_a} \partial_{\mu} \Phi - \frac{\delta F}{\delta \omega_a}$, and suppose $F(\Phi) = \Phi$

translation: $P_{\mu} = -i \partial_{\mu}$

dilation: $D = -i x^{\mu} \partial_{\mu}$ SCT: $K_{\mu} = -i(2x_{\mu} x^{\nu} \partial_{\nu} - x^2 \partial_{\mu})$

rotation: $L_{\mu\nu} = i(x_{\mu} \partial_{\nu} - x_{\nu} \partial_{\mu})$

Non-trivial commutation relation:

$$\begin{aligned}
 [D, P_\mu] &= i P_\mu \\
 [D, K_\mu] &= -i K_\mu \\
 [K_\mu, P_\nu] &= 2i (\eta_{\mu\nu} D - L_{\mu\nu}) \\
 [K_\rho, L_{\mu\nu}] &= i (\eta_{\rho\mu} K_\nu - \eta_{\rho\nu} K_\mu) \\
 [P_\rho, L_{\mu\nu}] &= i (\eta_{\rho\mu} P_\nu - \eta_{\rho\nu} P_\mu) \\
 [L_{\mu\nu}, L_{\rho\sigma}] &= i (\eta_{\nu\rho} L_{\mu\sigma} + \eta_{\mu\sigma} L_{\nu\rho} - \eta_{\mu\rho} L_{\nu\sigma} - \eta_{\nu\sigma} L_{\mu\rho})
 \end{aligned}$$

$$\begin{pmatrix} D & & & & \\ & P & & & \\ & & K & & \\ & & & L & \\ & & & & \ddots \end{pmatrix}$$

In order to simplify the commutation relation, we put:

$$J_{\mu\nu} = L_{\mu\nu} \quad J_{-1,\mu} = \frac{1}{2} (P_\mu - K_\mu) \quad J_{-1,0} = D \quad J_{0,\mu} = \frac{1}{2} (P_\mu + K_\mu)$$

$J_{\mu\nu}$ is antisymmetric, and obey $SO(d+1,1)$ commutation relation:

$$[J_{ab}, J_{cd}] = i (\eta_{ad} J_{bc} + \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac})$$

if spacetime metric is Euclidian

Generally, we have $SO(p, q) \xrightarrow{\text{map}} SO(p+1, q+1)$

conformal invariant:

the first 3 means invariant of type $\frac{|x_i - x_j|}{|x_k - x_m|}$

for SCT:

$$(x'_1 - x'_2)^2 = \left(\frac{x_1 - b x_1^2}{\Lambda_1^2} - \frac{x_2 - b x_2^2}{\Lambda_2^2} \right)^2$$

$$\Lambda_i^2 = 1 - 2b \cdot x_i + b^2 x_i^2 = \frac{x_1^2}{\Lambda_1^2} + \frac{x_2^2}{\Lambda_2^2} - 2 \frac{1}{\Lambda_1^2 \Lambda_2^2} (x_1 x_2 - x_1 b x_2^2 - x_2 b x_1^2 + b^2 x_1^2 x_2^2)$$

$$= \frac{1}{\Lambda_1^2 \Lambda_2^2} [x_1^2 \Lambda_2^2 + x_2^2 \Lambda_1^2 - 2(x_1 x_2 - x_1 b x_2^2 - x_2 b x_1^2 + b^2 x_1^2 x_2^2)]$$

$$\begin{aligned} \text{numerator} &= x_1^2 + x_2^2 - 2b \cdot x_2 x_1^2 + 2b^2 x_1^2 x_2^2 - 2b \cdot x_1 x_2^2 - 2x_1 \cdot x_2 + 2x_1 \cdot b x_2^2 + 2x_2 \cdot b x_1^2 - 2b^2 x_1^2 x_2^2 \\ &= x_1^2 + x_2^2 - 2x_1 \cdot x_2 \end{aligned}$$

$$\Rightarrow |x'_1 - x'_2| = \frac{|x_1 - x_2|}{\Lambda_1 \Lambda_2}$$

$$\text{So possible invariant: } \frac{|x_1 - x_2| |x_3 - x_4|}{|x_1 - x_3| |x_2 - x_4|} \quad (\text{simplest form})$$

Classical version:

new generator T_g : $\phi(x') = (1 - i\omega_g T_g) \phi(x)$, $\phi(x)$ is multicomponent field
for $F(\phi) \neq \phi$, there will be additional term in the algebra.

Consider Lorentz group first:

$$L_{\mu\nu} \phi(x) = S_{\mu\nu} \phi(x) \xrightarrow{\text{translate}} e^{ix^\rho P_\rho} L_{\mu\nu} e^{-ix^\rho P_\rho} = S_{\mu\nu} - x_\mu P_\nu + x_\nu P_\mu$$

$$\text{Root: } [P_\rho, L_{\mu\nu}] = i (\eta_{\rho\mu} P_\nu - \eta_{\rho\nu} P_\mu)$$

then we have:
$$\begin{cases} P_\mu \phi(x) = -i \partial_\mu \phi(x) \\ L_{\mu\nu} \phi(x) = i(x_\mu \partial_\nu - x_\nu \partial_\mu) \phi(x) + S_{\mu\nu} \phi(x) \end{cases}$$

Similarly, we can remove translation from the algebra

At $x=0$, $\begin{cases} L_{\mu\nu} \rightarrow S_{\mu\nu} \\ D \rightarrow \tilde{D} \\ K_\mu \rightarrow K_\mu \end{cases}$, then:

$$\begin{aligned} [\tilde{D}, K_\mu] &= -i K_\mu \\ [K_\rho, S_{\mu\nu}] &= i(\eta_{\rho\mu} K_\nu - \eta_{\rho\nu} K_\mu) \\ [S_{\mu\nu}, S_{\rho\sigma}] &= i(\eta_{\nu\rho} S_{\mu\sigma} + \eta_{\mu\sigma} S_{\nu\rho} - \eta_{\mu\rho} S_{\nu\sigma} - \eta_{\nu\sigma} S_{\mu\rho}) \end{aligned}$$

After translation:

$$\begin{cases} D \phi(x) = (\tilde{D} - i x^\nu \partial_\nu) \phi(x) \\ K_\mu \phi(x) = (K_\mu + 2 x_\mu \tilde{D} - 2 i x^\nu S_{\mu\nu} - 2 i x_\mu x^\nu \partial_\nu + i x^2 \partial_\mu) \phi(x) \end{cases}$$

if $\phi(x)$ is a irreducible representation of the Lorentz group,

$[\tilde{D}, S_{\mu\nu}] = 0$ Schur's Lemma $\tilde{D} = -i\Delta$

Notice: under global dialation, $x^\mu \rightarrow (1 + \delta a) x^\mu = e^{\delta a} x^\mu$ $\Rightarrow \begin{cases} x' = \lambda x \\ \phi'(x) = \lambda^{-\Delta} \phi(x) \end{cases}$

$\phi'(x) = \phi(x) (1 - i \delta a \cdot (-i\Delta)) = \phi(x) (1 - \Delta \delta a) = (e^{\delta a})^{-\Delta}$

Then we can define quasi-primary field:

$$\phi(x) \rightarrow \phi'(x) = \left| \frac{\partial x'}{\partial x} \right|^{-\Delta/\Delta} \phi(x)$$

Energy-momentum tensor

For symmetric $T_{\mu\nu}$,

$$\delta S = \int d^d x T^{\mu\nu} \partial_\mu \epsilon_\nu = \frac{1}{2} \int d^d x T^{\mu\nu} (\partial_\mu \epsilon_\nu + \partial_\nu \epsilon_\mu) = \frac{1}{2} \int d^d x T^{\mu\nu} \partial_\mu \epsilon_\nu$$

if $T^\mu_\mu = 0$ imply S is invariant under conformal transformation.

Scale invariance may imply traceless

For a generic field with scale invariance, $x'^\mu = (1 + \alpha) x^\mu$ $\mathcal{F}(\phi) = (1 - \alpha \Delta) \phi$:

$$j^\mu_\nu = x^\nu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \right] + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi = -\mathcal{L} x^\mu + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} x^\nu \partial_\nu \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi$$

$$\Rightarrow j^\mu_\nu = T^\mu_\nu x^\nu + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi \quad [\text{Hypothesis: } \partial_\mu j^\mu_\nu = 0]$$

Appendix: Belinfante Tensor [Chapter 2.5]

Canonically, $T^\mu_\nu = -\eta^{\mu\nu} \mathcal{L} + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial^\nu \phi \rightarrow$ generically not symmetric

Introduce Belinfante energy-moment tensor

$T_B^{\mu\nu} = T_c^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}$, where $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$

make $T_B^{\mu\nu} = T_B^{\nu\mu}$

Additional term doesn't affect classical conservation nor the Ward identity

Consider current associated with Lorentz transformation:

$$j^{\mu\nu\rho} = T_c^{\mu\nu} x^\rho - T_c^{\mu\rho} x^\nu + \frac{1}{2} i \frac{\partial \mathcal{L}}{\partial (\partial_\rho \phi)} S^{\nu\rho} \phi$$

If $j^{\mu\nu\rho} = T_B^{\mu\nu} x^\rho - T_B^{\mu\rho} x^\nu$, by $\begin{cases} \partial_\mu j^{\mu\nu\rho} = 0 \\ \partial_\mu T_B^{\mu\nu} = 0 \end{cases} \Rightarrow T_B^{\mu\nu} \delta^\rho_\mu - T_B^{\mu\rho} \delta^\nu_\mu = 0 \Leftrightarrow T_B^{\rho\nu} = T_B^{\nu\rho}$ symmetric!

By inspection, $B^{\mu\nu\rho} = \frac{1}{2} i \left[\frac{\partial L}{\partial(\partial_\mu \phi)} S^{\nu\rho} + \frac{\partial L}{\partial(\partial_\rho \phi)} S^{\mu\nu} + \frac{\partial L}{\partial(\partial_\nu \phi)} S^{\mu\rho} \right] \phi$

$S^{\mu\nu} = -S^{\nu\mu}$ ensure antisymmetry in the first 2 indices

$$\begin{cases} \partial^\mu \rho^\nu - B^{\mu\nu\rho} = \frac{1}{2} i \frac{\partial L}{\partial(\partial_\mu \phi)} S^{\nu\rho} \phi \\ \partial_\mu j^{\mu\nu\rho} = 0 \Leftrightarrow T_c^{\rho\nu} - T_c^{\nu\rho} = -\frac{1}{2} i \partial_\mu \left[\frac{\partial L}{\partial(\partial_\mu \phi)} S^{\nu\rho} \phi \right] = \partial_\mu B^{\mu\nu\rho} - \partial_\mu B^{\mu\rho\nu} \end{cases}$$

Check the expression of $j^{\mu\nu\rho}$!

Define: $v^\mu = \frac{\partial L}{\partial(\partial^\rho \phi)} (\eta^{\mu\rho} \partial + i S^{\mu\rho}) \phi$ Assuming $v^\mu = \partial_\alpha \sigma^{\alpha\mu}$

Then define: $\sigma_+^{\mu\nu} = \frac{1}{2} (\sigma^{\mu\nu} + \sigma^{\nu\mu})$

$$X^{\lambda\rho\mu\nu} = \frac{2}{d-2} \left\{ \eta^{\lambda\rho} \sigma_+^{\mu\nu} - \eta^{\lambda\mu} \sigma_+^{\rho\nu} - \eta^{\lambda\nu} \sigma_+^{\rho\mu} + \eta^{\mu\nu} \sigma_+^{\lambda\rho} + \frac{1}{d-1} (\eta^{\lambda\rho} \eta^{\mu\nu} - \eta^{\lambda\mu} \eta^{\rho\nu}) \sigma_+^\alpha{}_\alpha \right\}$$

$$\text{consider } T^{\mu\nu} = \underbrace{T_c^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}}_{\text{Belinfante}} + \frac{1}{2} \partial_\lambda \partial_\rho X^{\lambda\rho\mu\nu}$$

traceless

Not check at all $\leftarrow$ For $X^{\lambda\rho\mu\nu}$: (1) $\partial_\mu \partial_\lambda \partial_\rho X^{\lambda\rho\mu\nu} = 0 \rightarrow$ not affect conservation

$$(2) X^{\lambda\rho\mu\nu} - X^{\lambda\rho\nu\mu} = \frac{2}{(d-2)(d-1)} \sigma_+^\alpha{}_\alpha (\eta^{\lambda\mu} \eta^{\rho\nu} - \eta^{\lambda\nu} \eta^{\rho\mu})$$

$\partial_\lambda \partial_\rho (\dots) = 0 \Leftrightarrow$ Not spoil the symmetry of Belinfante tensor

$$\begin{aligned} (3) \frac{1}{2} \partial_\lambda \partial_\rho X^{\lambda\rho\mu}{}_\mu &= \partial_\mu v^\mu, \quad \partial_\rho B^{\rho\mu}{}_\mu = \frac{1}{2} i \partial_\rho \left(\frac{\partial L}{\partial(\partial_\rho \phi)} S^{\mu\rho} \phi \right) \\ &\Rightarrow T^\mu{}_\mu = T_c^\mu{}_\mu + \Delta \partial_\mu \left(\frac{\partial L}{\partial(\partial_\mu \phi)} \phi \right) = \partial_\mu j^\mu = 0 \end{aligned}$$

In a nutshell, scale invariant and $v^\mu = \partial_\alpha \sigma^{\alpha\mu}$ imply the existence of traceless $T^{\mu\nu}$

We hope the $T^{\mu\nu}$ in $d=2$ still holds, in quantum version, the vacuum expectation of $(T^\mu{}_\mu)^2$ vanish in $d=2$ if conformal invariance is presented.

Quantum version: mainly focus of conformal invariance on two/three point correlation function of **quasi-primary** fields.

"field": $\phi, \partial_\mu \phi, \dots \rightarrow$ local quantities

consider spinless field:

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \left| \frac{\partial x'_1}{\partial x} \right|_{x'_1=x_1}^{\Delta_1/d} \left| \frac{\partial x'_2}{\partial x} \right|_{x'_2=x_2}^{\Delta_2/d} \langle \phi_1(x'_1) \phi_2(x'_2) \rangle$$

for scale transformation $x \rightarrow \lambda x$

for rotation & translation

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \lambda^{\Delta_1+\Delta_2} \langle \phi_1(\lambda x_1) \phi_2(\lambda x_2) \rangle = f(x_1 - x_2) = \lambda^{\Delta_1+\Delta_2} f(\lambda(x_1 - x_2))$$

$$\Leftrightarrow f(x) = \lambda^{\Delta_1+\Delta_2} f(\lambda x) \Rightarrow \frac{f(x)}{\lambda^{\Delta_1+\Delta_2+1}} (-\Delta_1 - \Delta_2) = x f'(x)$$

$$\Rightarrow f'(x) / f(x) = -(\Delta_1 + \Delta_2) \frac{1}{x} \Rightarrow f(x) = \frac{C_n}{x^{\Delta_1+\Delta_2}}$$

$$\langle \phi_1(x_1) \phi_2(x_2) \rangle = \frac{C_n}{|x_1 - x_2|^{\Delta_1+\Delta_2}}$$

for SCT: $|\frac{\partial x'}{\partial x}| = \frac{1}{\Lambda^2 a}$

$$\Rightarrow \langle \phi_1(x_1) \phi_2(x_2) \rangle = \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}} = \frac{1}{\Lambda^{\Delta_1} \Lambda^{\Delta_2}} (\Lambda_1 \Lambda_2)^{\Delta_1 + \Delta_2} \frac{C_{12}}{|x_1 - x_2|^{\Delta_1 + \Delta_2}} \quad (\lambda = \Lambda^2)$$

4.56 Not check yet! $\Rightarrow \Delta_1 = \Delta_2$
 $C_{12} = 0$, $\langle \phi_1(x_1) \phi_2(x_2) \rangle = \begin{cases} 0 & \Delta_1 \neq \Delta_2 \\ \frac{C_{12}}{|x_1 - x_2|^{2\Delta}} & \Delta_1 = \Delta_2 = \Delta \end{cases}$

Then for 3-point functions, similarly rotation, translation, dialation give:

$$\langle \phi_1 \phi_2 \phi_3 \rangle = \frac{C_{123}^{(a,b,c)}}{x_{12}^a x_{23}^b x_{13}^c} \quad a+b+c = \Delta_1 + \Delta_2 + \Delta_3$$

for SCT: $\frac{1}{y_1^{\Delta_1} y_2^{\Delta_2} y_3^{\Delta_3}} \frac{1}{x_1^{\frac{a+c}{2}} x_2^{\frac{a+b}{2}} x_3^{\frac{b+c}{2}}} = 1$

the result may require the correlation to be analytic

$$\Rightarrow \begin{cases} a = \Delta_1 + \Delta_2 - \Delta_3 \\ b = \Delta_2 + \Delta_3 - \Delta_1 \\ c = \Delta_3 + \Delta_1 - \Delta_2 \end{cases} \Rightarrow \langle \phi_1 \phi_2 \phi_3 \rangle = \frac{C_{123}}{x_{12}^{\Delta_1 + \Delta_2 - \Delta_3} x_{23}^{\Delta_2 + \Delta_3 - \Delta_1} x_{13}^{\Delta_3 + \Delta_1 - \Delta_2}}$$

For more point function, it's no more elegant!

Ward Identities: Consequence of a symmetry of the action and the measure on correlation.

Appendix [Chapter 2.4.3 & 2.4.4]

Consider a correlation function $\langle \phi(x_1) \dots \phi(x_n) \rangle = \frac{1}{Z} \int [d\phi] \phi(x_1) \dots \phi(x_n) e^{-S[\phi]}$

Consequence of symmetry $S[\phi]$ invariant

$$\hookrightarrow \langle \phi(x_1) \dots \phi(x_n) \rangle = \langle F[\phi(x_1)] \dots F[\phi(x_n)] \rangle, *$$

Besides, infinitesimal transformation gives $\phi'(x) = \phi(x) - i\omega_a G_a \phi(x)$

$$\langle X \rangle = \frac{1}{Z} \int [d\phi] \phi(x) \exp \left\{ -S[\phi] + \int dx \partial_\mu j_a^\mu \omega_a(x) \right\}$$

Expand to the first order of $\omega_a(x) \rightarrow \langle \delta X \rangle = \int dx \partial_\mu \langle j_a^\mu(x) X \rangle \omega_a(x)$

$$\text{However, } \delta X = -i \left[\sum_{i=1}^n \phi(x_i) \dots G_a \phi(x_i) \dots \phi(x_n) \right] \omega_a(x_i)$$

$$= -i \int dx \omega_a(x) \left[\sum_{i=1}^n \phi(x_i) \dots G_a \phi(x_i) \dots \phi(x_n) \delta(x - x_i) \right]$$

$$\Rightarrow \frac{\partial}{\partial x^\mu} \langle j_a^\mu(x) \phi(x_1) \dots \phi(x_n) \rangle = -i \sum_{i=1}^n \langle \phi(x_1) \dots G_a \phi(x_i) \dots \phi(x_n) \rangle \delta(x - x_i)$$

[Ward identity for the current j_a^μ]

Integral gives:

$$\delta_\omega X = -i \omega_a \sum_{i=1}^n \langle \phi(x_i) \dots G_a \phi(x_i) \dots \phi(x_n) \rangle = 0 \rightarrow \text{infinite version of } (*)$$

The Ward identity of translation symmetry:

$$\partial_\mu \langle T^{\mu\nu} X \rangle = - \sum_i \delta(x - x_i) \frac{\partial}{\partial x_i^\nu} \langle X \rangle, \text{ this equation still holds after modification}$$

Consider Lorentz (Rotation) symmetry: $j^{\mu\nu\rho} = T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu$

the ward identity gives

$$\partial_\mu \langle (T^{\mu\nu} x^\rho - T^{\mu\rho} x^\nu) X \rangle = \sum_i \delta(x - x_i) [x_i^\nu \partial_i^\rho - x_i^\rho \partial_i^\nu \langle X \rangle - i S_i^{\nu\rho} \langle X \rangle]$$

substitute

$$\Rightarrow \langle (T^{\rho\nu} - T^{\nu\rho}) \chi \rangle = -i \sum_i \delta(x-x_i) S_i^{\rho\nu}(x)$$

Notice: When $x=x_i$, $\langle (T^{\rho\nu} - T^{\nu\rho}) \chi \rangle$ generically non-zero

Consider scale invariance: $j_0^\mu = T^\mu_\nu x^\nu$ ($T^{\mu\nu}$ symmetric and traceless)

$$\partial_\mu \langle T^\mu_\nu x^\nu \chi \rangle = - \sum_i \delta(x-x_i) \{ x_i^\nu \frac{\partial}{\partial x_i^\nu} \langle \chi \rangle + \Delta_i \langle \chi \rangle \}$$

$$\Rightarrow \langle T^\mu_\mu \chi \rangle = - \sum_i \delta(x-x_i) \Delta_i \langle \chi \rangle$$

$T_{\mu\nu}$ in two dimensions

Consider two-point function of energy-momentum tensor
(Schwinger function)

$$S_{\mu\nu\rho\sigma}(x) = \langle T_{\mu\nu}(x) T_{\rho\sigma}(0) \rangle \quad T_{\mu\nu} \text{ is symmetric and conserved}$$

$$S_{\mu\nu\rho\sigma} = S_{\nu\mu\rho\sigma} = S_{\nu\mu\sigma\rho} = S_{\mu\nu\sigma\rho} \leftarrow$$

$$\text{Translation symmetry: } S_{\mu\nu\rho\sigma}(x) = \langle T_{\mu\nu}(0) T_{\rho\sigma}(x) \rangle = \langle T_{\rho\sigma}(0) T_{\mu\nu}(x) \rangle = S_{\rho\sigma\mu\nu}(x)$$

$$\text{Moreover, parity gives: } S_{\mu\nu\rho\sigma}(x) = S_{\rho\sigma\mu\nu}(x)$$

$$\text{Scale invariance gives: } S_{\mu\nu\rho\sigma}(\lambda x) = \lambda^{-4} S_{\mu\nu\rho\sigma}(x)$$

May come from normal ordering

The general solution gives:

Not check yet! $\leftarrow S_{\mu\nu\rho\sigma}(x) = (x^2)^{-4} \{ A_1 g_{\mu\nu} g_{\rho\sigma} (x^2)^2 + A_2 (g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho}) (x^2)^2 \right.$

$$\left. + A_3 (g_{\mu\nu} x_\rho x_\sigma + g_{\rho\sigma} x_\mu x_\nu) x^2 + A_4 x_\mu x_\nu x_\rho x_\sigma \right\}$$

$$\text{because } \partial^\mu T_{\mu\nu} = 0 \Rightarrow \partial^\mu S_{\mu\nu\rho\sigma}(x) = 0$$

$$\Rightarrow A_1 = 3A \quad A_2 = -A \quad A_3 = -4A \quad A_4 = 8A$$

$$\Rightarrow S_{\mu\nu\rho\sigma}(x) = \frac{A}{(x^2)^4} \{ (3g_{\mu\nu} g_{\rho\sigma} - g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}) (x^2)^2 - 4x^2 (g_{\mu\nu} x_\rho x_\sigma + g_{\rho\sigma} x_\mu x_\nu) + 8x_\mu x_\nu x_\rho x_\sigma \}$$

$$\text{then } S^\mu{}_\mu{}^\sigma{}_\sigma(x) = \frac{A^2}{(x^2)^4} \{ [3 \times 2 \times 2 - 2 - 2 - 4 \times (2+2) + 8] (x^2)^2 \} = 0$$

$$\text{especially, } S^\mu{}_\mu{}^\sigma{}_\sigma(0) = \langle T^\mu{}_\mu(0) \rangle = 0$$